Case 1 – Thredz

Thredz is a successful costume hire business. Its customers range from theatre companies, who hire large numbers of outfits for periods of several weeks, to individuals who hire a single costume for one day for a fancy dress party.

Details of the costumes available for hire, such as description and daily hire rate are held on a PC system. Thredz will hold a number of examples of each costume (some in differing sizes), with each individual item for hire being identified by a unique number held on a barcode label. Details of costumes and the individual barcoded items are added to the PC system as soon as Thredz receives items from their manufacturers.

Each costume is allocated to one or more "themes", such as Christmas, Halloween, The Romans, and Weddings etc. The staff at Thredz uses these themes in order to answer customer enquiries about what costumes they have available for specified occasions.

When customers wish to hire an item, its code is entered into the PC system along with the required hire period. The hire period may start immediately and run for a number of days, or begin and end on specified dates in the future.

The system then checks that the item has not been hired already for all or part of the requested hire period. If it has been hired already, then alternative items are suggested, and the customer is given the choice of cancelling the hire, changing the hire period or selecting an alternative item.

Once all of the items that the customer wishes to hire have been entered into the system, details of the customer are entered, or recalled from the customer database if the customer has hired a costume previously. The system calculates the fee for all of the items to be hired and produces a hire agreement, detailing each item, its hire period and its individual hire fee, which the customer signs to signify that they agree with the terms and conditions of the hire. Not all of the items on an agreement will necessarily share the same hire period. Some items may be taken immediately and others collected at a later date. They may also be returned at different times.

As each item is issued to the customer, its code is again entered into the system, which records the item as being out on hire. At this point the customer is charged a returnable deposit equal to the value of the item. As each item is returned, its code is entered into the system along with the customer's hire agreement number. If an item has been damaged a charge may be applicable up to the value of the item. An invoice is produced for all items returned at the same time (they may have been hired on different occasions using different hire agreements), payment is taken from the customer, and the deposits are refunded.

Case 2 – Fresco

Fresco is a ticket agency, dealing in concert and theatre tickets. All of their business is conducted over the telephone, with customers ringing up to request tickets for a wide variety of performances.

Concert and theatre venues provide Fresco with a constant stream of information on forthcoming events, which is then used by Fresco's manager to compile a fixture list for use by the sales staff in responding to customer calls. The manager will also select a number of events for which Fresco will purchase tickets in advance of customer requests (e.g. for popular events). Details of these 'pre-purchase' orders as they are known are passed to the post clerk who is responsible for placing orders with each venue. The post clerk sends out each order with its payment attached. Once tickets are received the post clerk files them in the tickets file.

When customers ring the sales team their ticket requests are checked against the ticket file. If pre-purchased tickets are available they are put in an envelope marked with the customer's name and filed at the back of the ticket file. If not, the sales team fill out a ticket request form and put it in a desk tray for collection by the post clerk, who will then place an order with the appropriate venue in the same way as for pre-purchased tickets.

Details of the customer and the tickets are passed to the payments section for invoicing (by the sales team for pre-purchased tickets and by the post clerk for new tickets). The payments section will then send an invoice to the customer, and await payment if the customer is paying by post or accept payment immediately for credit card holders. A copy of the invoice is placed in the invoice file for matching with payments, and once paid a further copy is placed in a tray labeled 'dispatch list' for collection by the post clerk.

The post clerk collects paid invoice copies three times a day, and retrieves the appropriate tickets from the ticket file for dispatch to customers.

Case 3 – Natlib

Natlib is a small private library specializing in natural history books. They have a collection of titles available for loan by registered readers, free of charge.

Each reader is allowed to borrow up to eight books at a time. Loans are to be recorded against particular book copies, rather than their title, as Natlib may have several copies of any given book.

When all copies of a book are already out on loan a reader may wish to place a reservation for it. Each reservation is recorded against a book title. When a copy of the title is subsequently returned Natlib will place it on one side, record which copy is to satisfy the reservation, and notify the reader that it is ready for collection

The new system will cover functions in several areas: reader registration; book registration; book loans; book reservations; book returns; and issue of loan reminders.

When readers borrow a book their status is checked to see if they have already reached their loan limit or have any outstanding overdue loans or fines, in which case they will not be allowed to borrow any more books until they have taken appropriate action.

New loans will be recorded and details of their 'due for return' date given to the reader. Overdue loans will be monitored and up to three reminders sent to the reader. Each time a reminder is sent its date will be recorded against the reader, whose status will then be updated.

When readers reserve a book the title will be checked to ensure that Natlib currently holds it, and if so the reservation will be recorded.

When a reader returns a book its return date will be recorded against the loan record and its condition (if it has changed) against its copy record. For overdue returns and damaged copies the librarian will levy a fine against the reader and record its amount (and any subsequent) payment on the loan record. If payment is not made in full, the reader's status will be updated.

Each return will be checked against outstanding reservations. If the title is reserved the book copy will be placed on one side, the reservation will be updated with details of which copy has been returned and the appropriate reader will be notified.

The actual purchase of new books will not be covered by the new system as it already functions well. However the system will need to record details of new titles, as sent in by book suppliers and publishers, and of new copies as the librarians at Natlib register them.